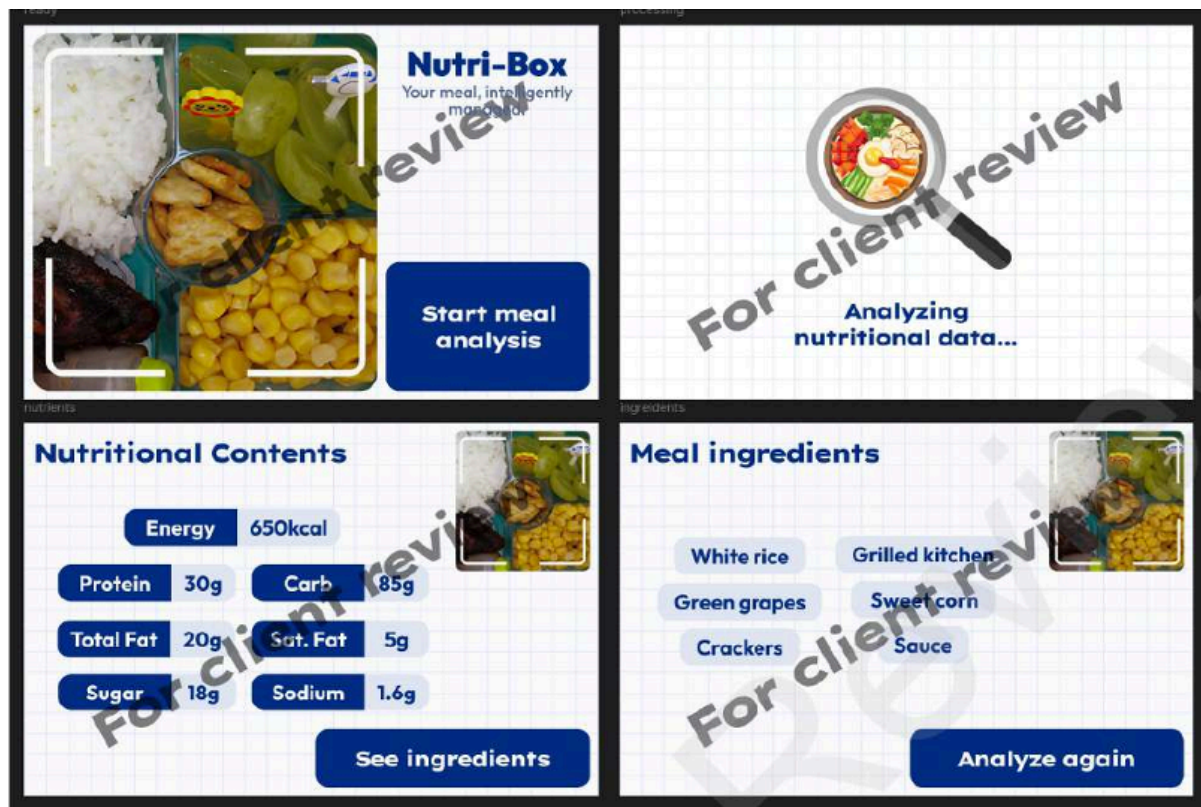


Sample Screen

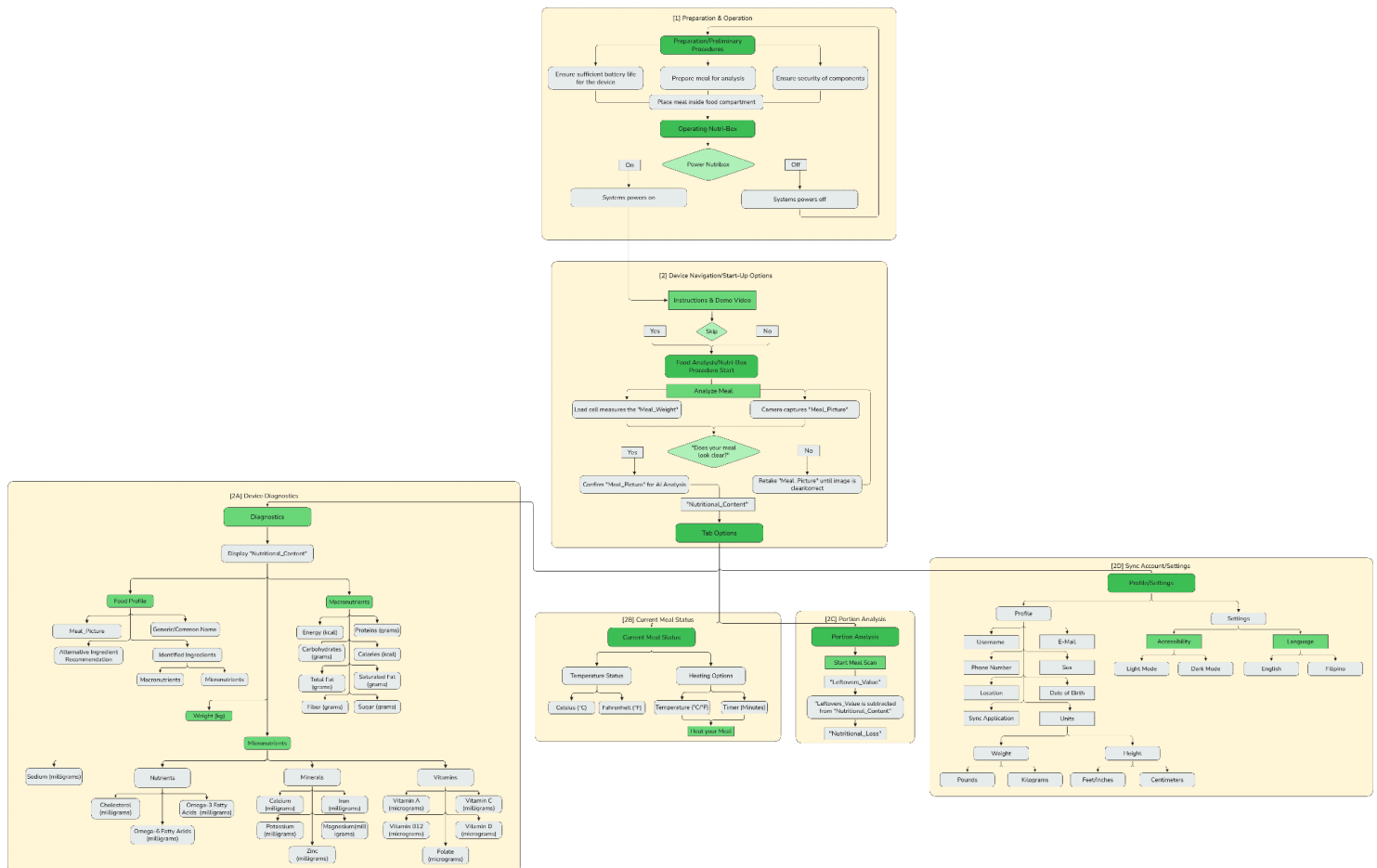


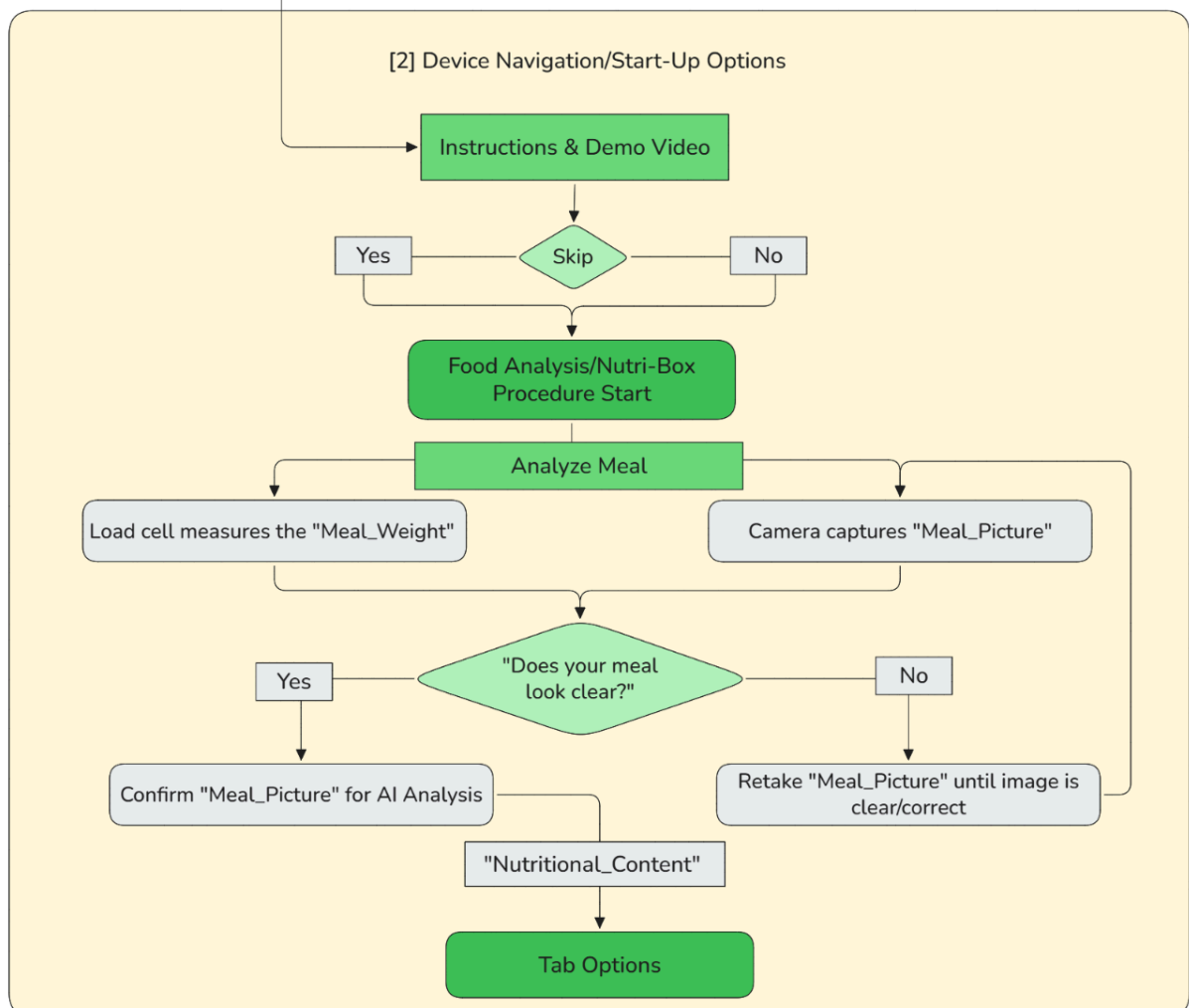
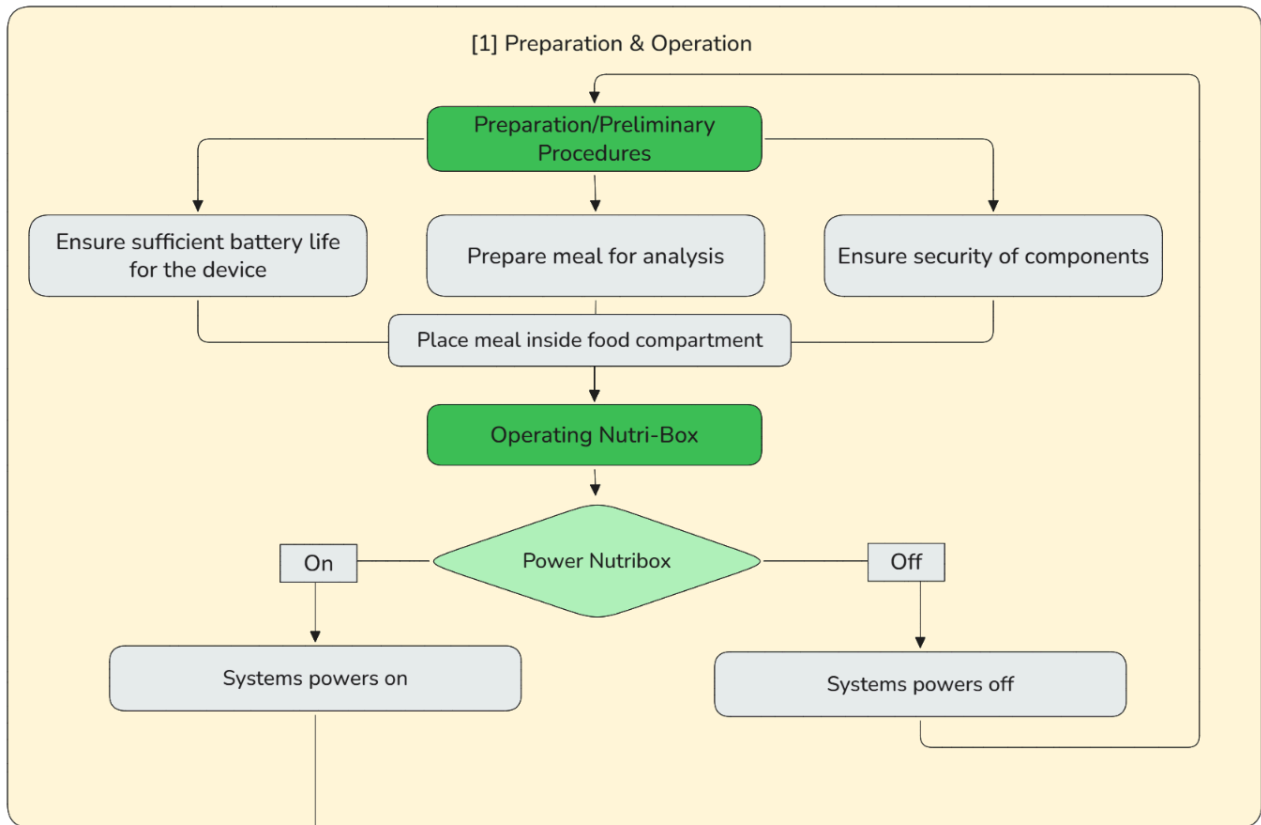
Device Features

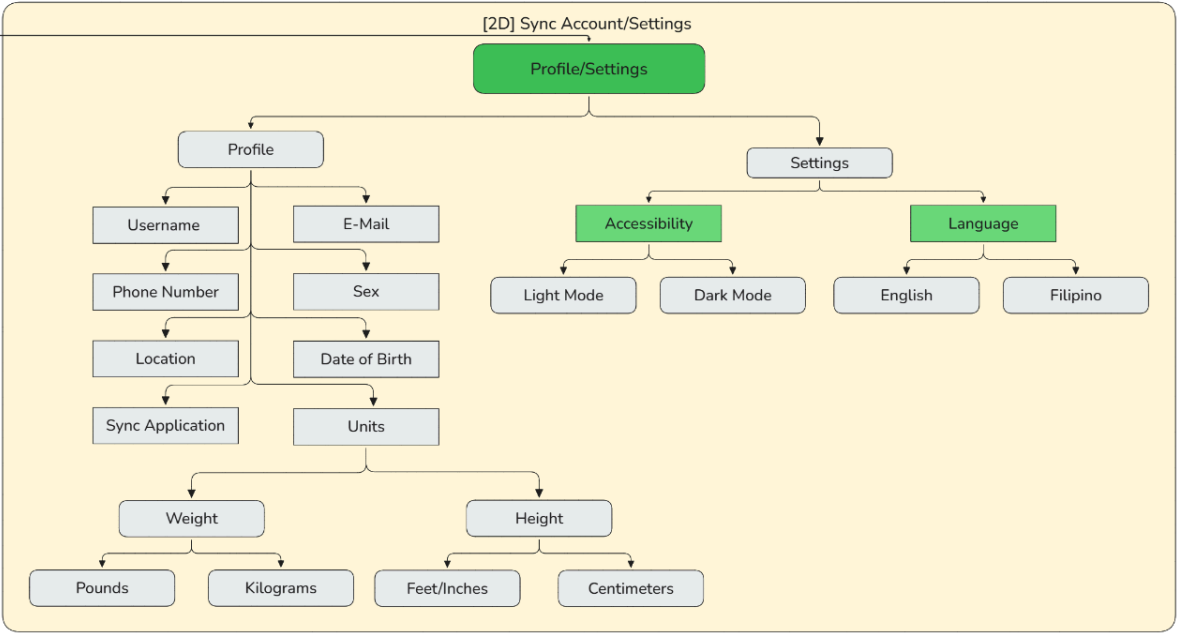
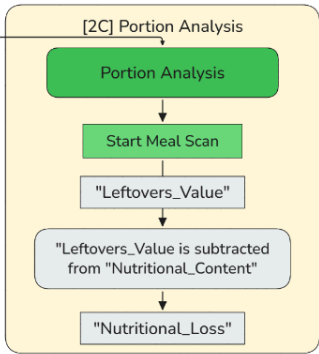
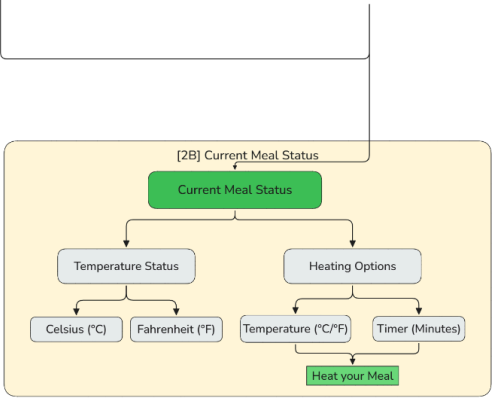
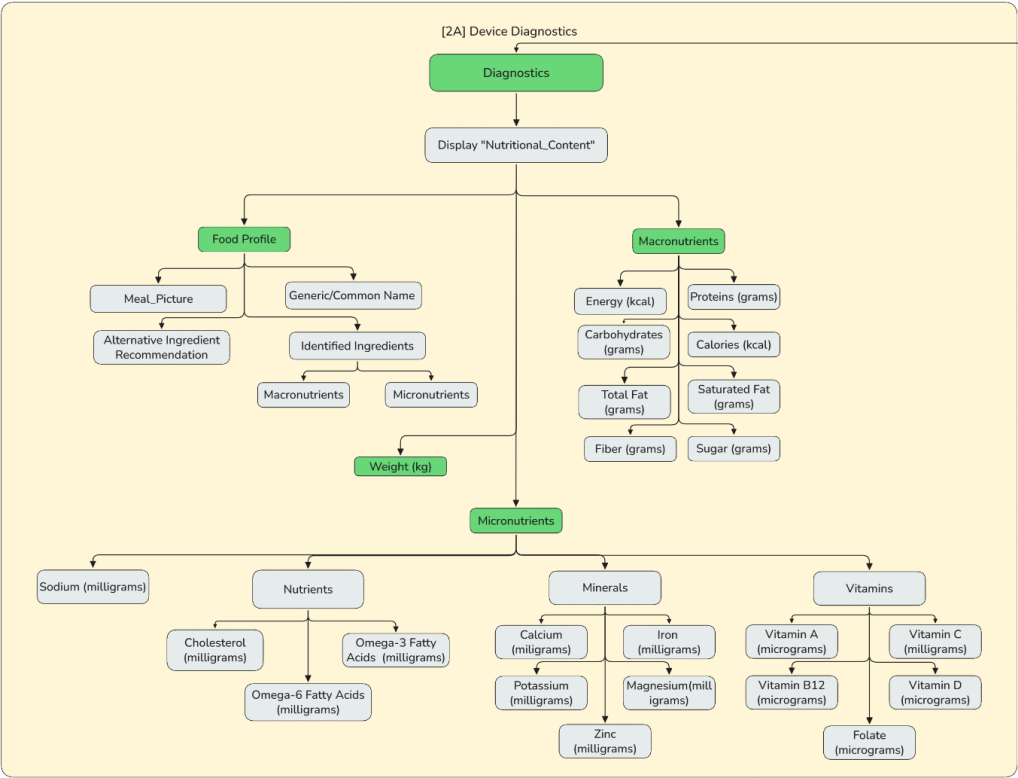
Feature	Description / Functionality
Nutri-Box 1.0 Baseline	Retains all existing functional requirements and features from Nutri-Box 1.0.
Image Capture	Captures a top-down digital image of the food inside the device using the camera mounted on the lid.
Food Recognition	Sends the captured food image to an external AI model through an API to identify the food items.
Weight Measurement	Measures the total weight of the meal using a load cell sensor and onboard scale.
Temperature Sensing	Measures and displays the current temperature of the meal using a temperature sensor.
Meal Heating / Reheating	Reheats the meal using a heating pad at selectable temperatures of 60°C, 65°C, 70°C, 75°C, or 80°C for 5, 10, 15, or 20 minutes .
Nutrient Estimation	Sends the identified food list and measured meal weight to the AI model to estimate the meal's nutritional content.

Nutrient Estimation Scope	Estimates energy, protein, carbohydrates, total fat, saturated fat, dietary fiber, sugars, and sodium , along with additional micronutrients supported by the system.
Food Database Integration	Uses nutritional databases, including the USDA database and locally developed databases, to match recognized foods with nutritional information.
Ingredient Identification	Identifies individual ingredients through an AI/automatic mode or allows the user to manually search for an ingredient.
Alternative Food Recommendation	Generates alternative ingredient or food recommendations based on the analyzed meal and the user's nutritional needs.
Data Display	Displays recognized foods, nutritional values, meal status, temperature status, diagnostics, portion analysis, and other system information on the device.
Post-Scan / Portion Analysis	Re-scans leftover food after consumption and compares it with the original meal to calculate nutritional loss.

Flowchart







Sample Screen

9:41

Profiling

Age:

Weight: pounds

Height: inches

Sex: ☒ Female ☐ Male

Ethnicity:

Location:

Dietary Restrictions

something something blabla

Allergies

Shellfish

Nuts

Medical Needs

Medical Condition

Lifestyle diets

Pregnancy

Other

Months

Drinking History

Smoking History

Medical Metrix

blablablablablabamlabnsnnblablabla
mlabsnsnn consult with professional

Body type

Blood type

Level of physical activity

Sedentary

Lightly Active

Moderately Active

Very Active

Highly Active

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Budget Allotment

blablababamlabnsnn
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blablababamlabnsnn
blablababamlabnsnn

Goal

Create Profile

9:41

nutribox

Username

Email/Phone Number

Password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Sign Up

9:41

Medical Metrix

blablablablablabamlabnsnnblablabla
mlabsnsnn consult with professional

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Budget Allotment

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Goal

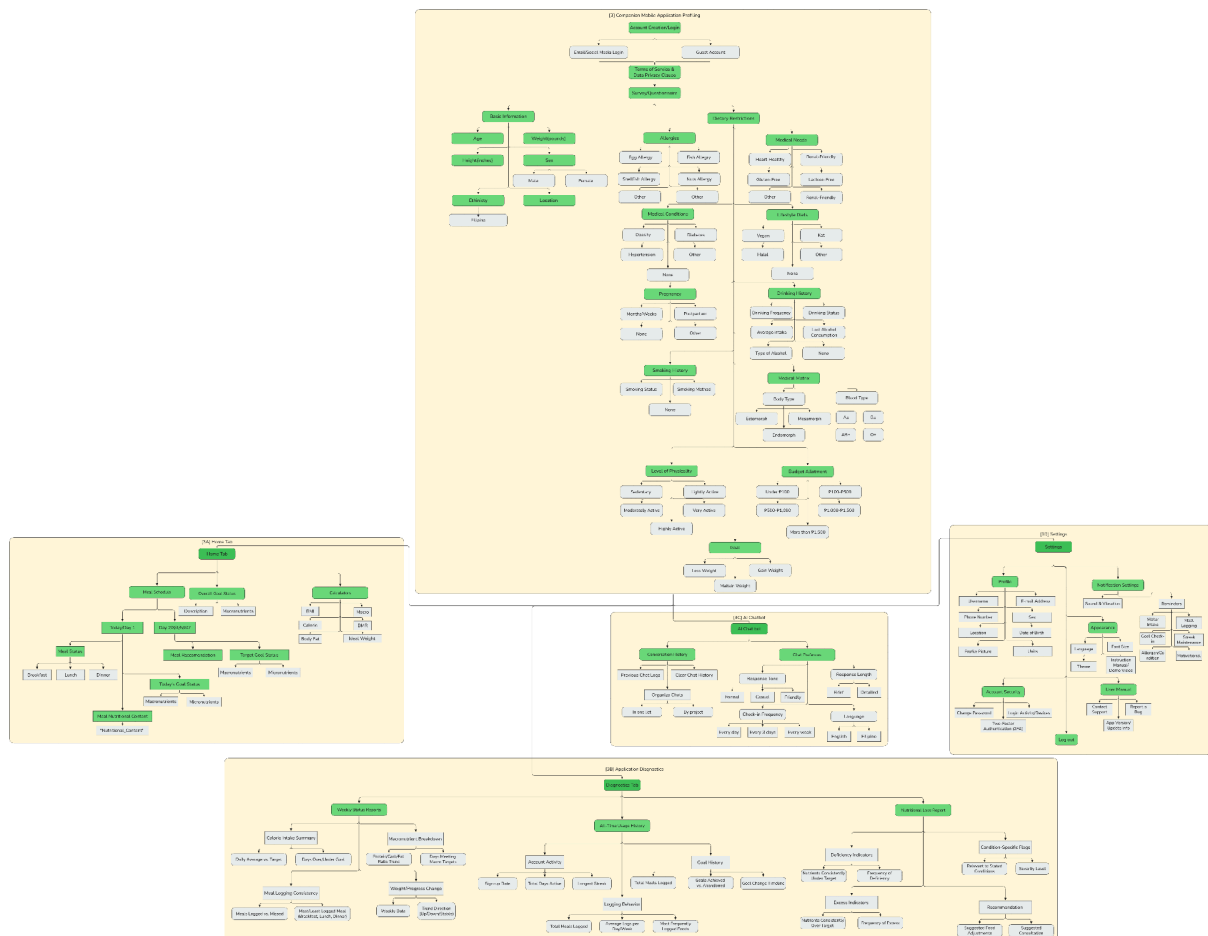
Create Profile

Application Features

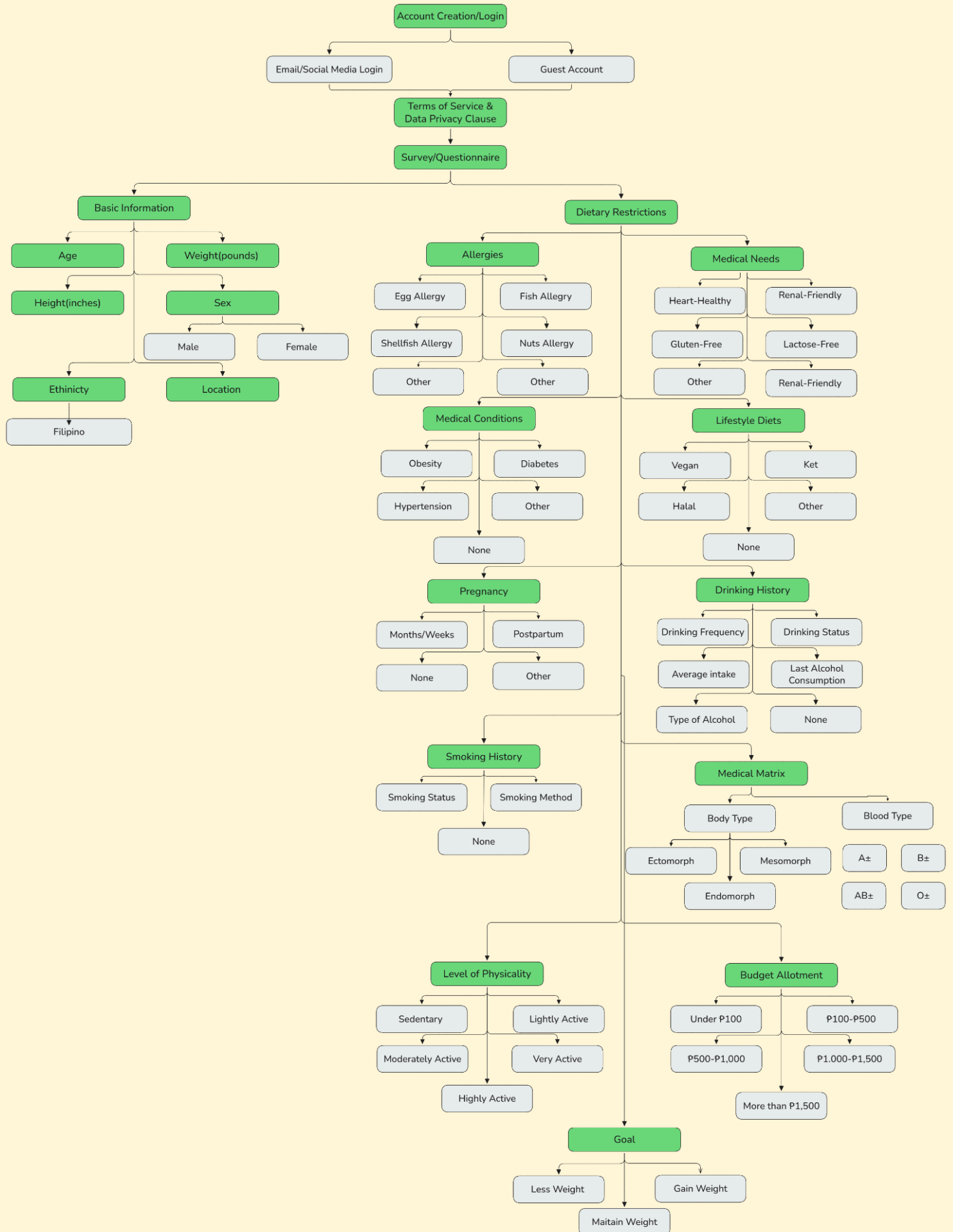
Companion Mobile Application	Provides the main mobile application interface containing Profiling, Home, Diagnostics, AI Chatbot, and Settings.
User Onboarding / Profiling	Provides a one-time setup process for entering the user's personal, dietary, health, activity, and nutritional information.
Profile Data Scope	Collects information such as health goals, dietary restrictions, allergies, medical conditions, lifestyle diet, physical activity, and other profile information.
Local Profile Storage	Securely stores collected profile information locally so it can be used for personalization.
Personalized Recommendations	Sends the user's profile, recognized foods, and estimated nutritional values to the AI model to generate personalized healthier food alternatives.
Recommendation Display	Displays recommended healthier alternatives and identifies which original food item each recommendation is intended to replace.
Nutrition Tracking	Records meal weight, meal time, identified foods, macronutrients, micronutrients, and total nutritional values in the user's dietary log.
Daily Nutrition Dashboard	Shows current nutritional intake, meals for the day, target intake, and macronutrient progress.
Meal Scheduling	Provides planned nutritional intake and meal information for upcoming meals and continuously updates the schedule.
Nutrition Calculators	Includes Calorie Needs/BMR/TDEE, Macro Split, Water Intake, Portion Size, EER, and BMI calculators.
Diagnostics	Provides weekly reports covering calorie intake, macronutrient trends, meal logging consistency, weight/progress changes, and usage history.
Nutritional Loss Report	Identifies nutrient deficiencies, excessive nutrient intake, condition-specific concerns, and provides suggested food adjustments.
AI Chatbot	Provides an AI-based assistant for nutrition-related questions, guidance, and personalized assistance.

Chat Personalization	Allows users to customize response tone, response length, language, guidance style, explanation depth, and proactive suggestions.
Notifications	Provides reminders, allergen warnings, goal-tracking notifications, hardware status updates, and other system notifications.
Account Management	Supports account registration/login, profile management, password changes, two-factor authentication, linked accounts, and account deletion.
Appearance & Settings	Allows users to configure theme, font size, language, accent color, layout, and other application settings.
Device Connectivity	Supports Wi-Fi/mobile hotspot connectivity and synchronization between the Nutri-Box and mobile application.
Database Management	Maintains USDA, local food, and school canteen recipe databases with nutritional information, ingredients, portion sizes, allergens, and menu schedules.

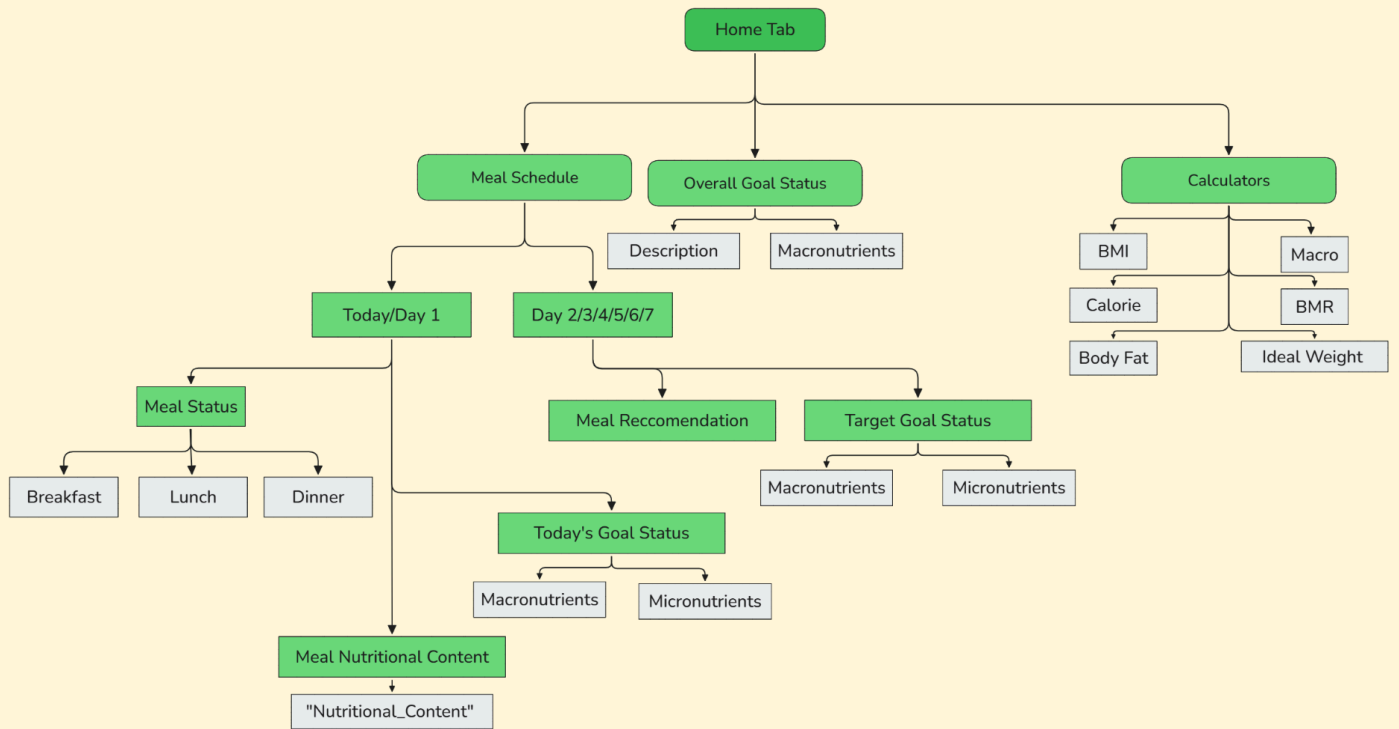
Flowchart



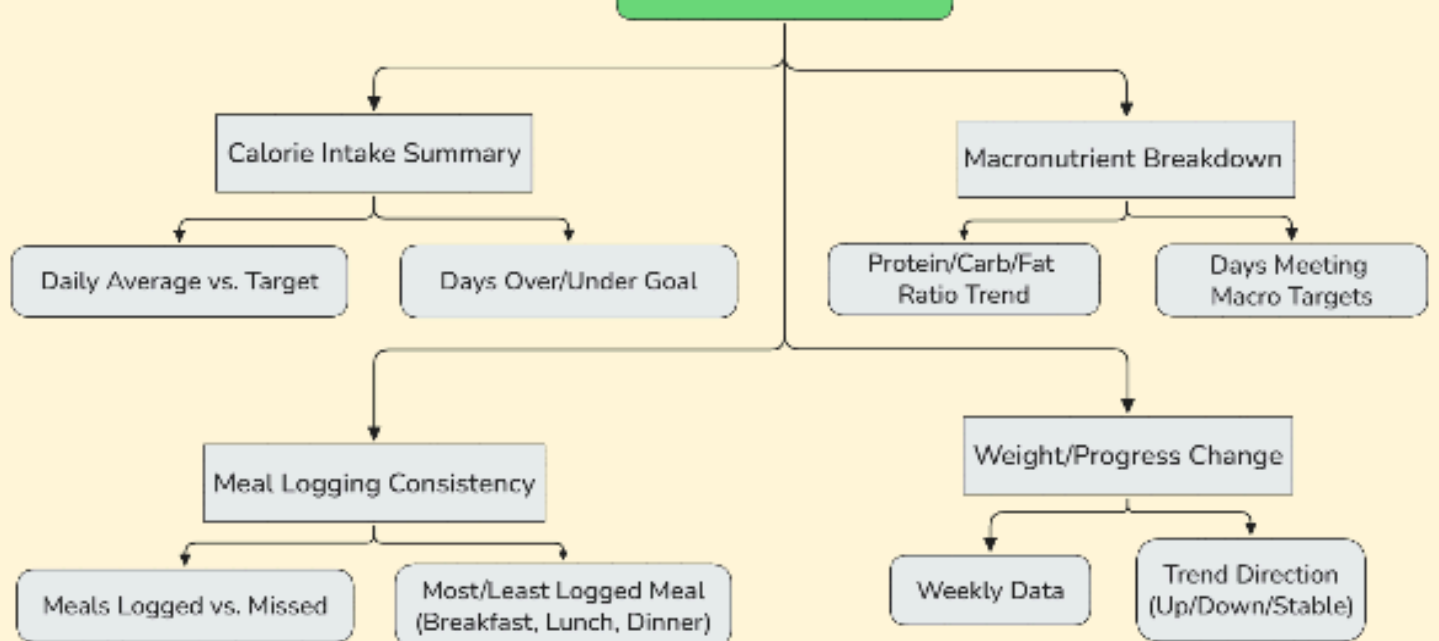
[3] Companion Mobile Application Profiling



[3A] Home Tab



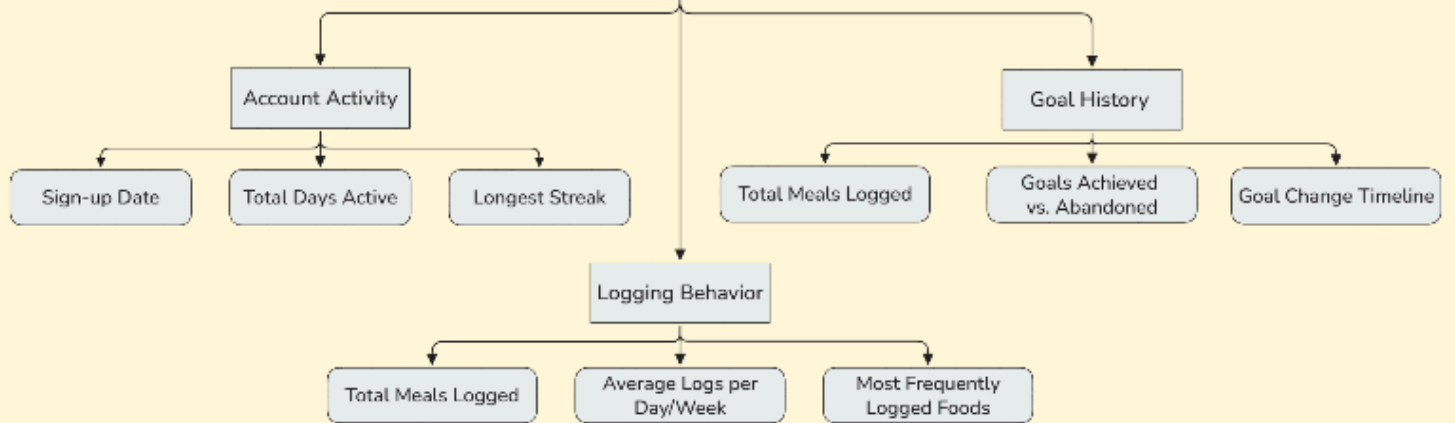
Weekly Status Reports



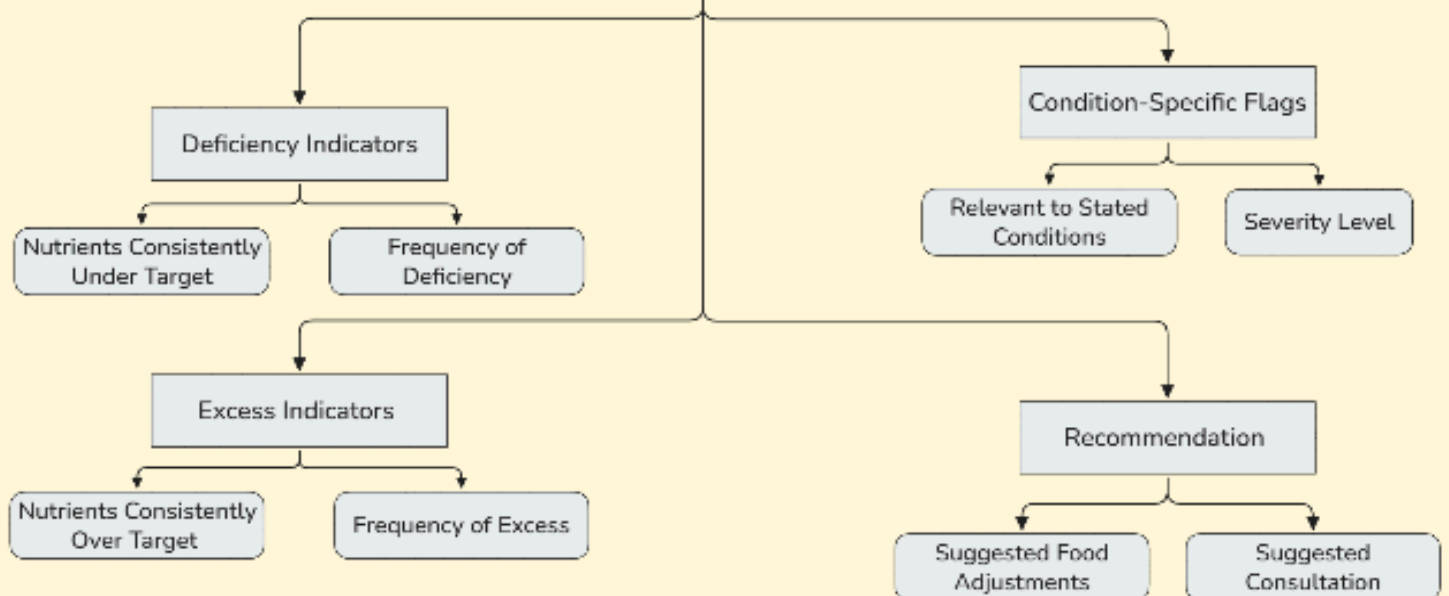
[3B] Application Diagnostics

Diagnostics Tab

All-Time Usage History

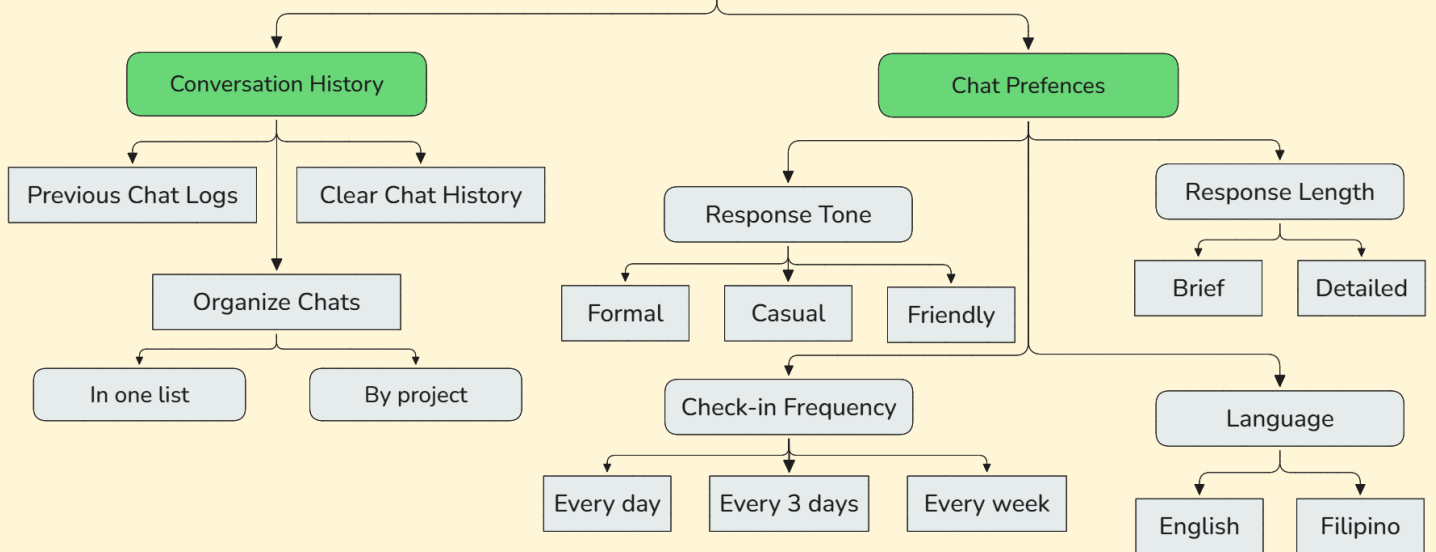


Nutritional Loss Report



[3C] AI Chatbot

AI Chat bot



[3D] Settings

Settings

